

The Gale Encyclopedia of Medicine (Second Edition) - Comprehensive Summary

Part 1: Overview and Structure

Publication Information

The Gale Encyclopedia of Medicine, Second Edition is a comprehensive medical reference published in 2002 by Gale Group. Edited by Jacqueline L. Longe with Associate Editor Deirdre S. Blanchfield, this encyclopedia is organized across five volumes covering nearly 1,700 medical topics.

Volume Organization

- Volume 1: Entries A-B (pages 1-624)
- Volume 2: Entries C-F (pages 625-1374)
- Volume 3: Entries G-M (pages 1375-2306)
- Volume 4: Entries N-S (pages 2307-3236)
- Volume 5: Entries T-Z (pages 3237-3602), plus Organizations directory (pages 3603-3624) and General Index (pages 3625+)

Purpose and Scope

This encyclopedia serves as a medical reference designed to inform and educate readers about disorders, conditions, treatments, and diagnostic tests. It fills the gap between basic consumer health resources and highly technical professional materials, using language that laypersons can understand while providing thorough coverage.

Content Coverage

The encyclopedia includes approximately 1,700 full-length articles covering:

- Disorders and conditions
- Medical tests and procedures
- Treatments and therapies
- Common drugs (with generic and brand names)
- Alternative medicine and holistic therapies
- Biographical sidebars of prominent medical figures

Part 2: Article Structure and Format

Standardized Article Format

For Disorders/Conditions:

- Definition
- Description
- Causes and symptoms
- Diagnosis
- Treatment
- Alternative treatment
- Prognosis
- Prevention
- Resources
- Key terms

For Tests/Treatments:

- Definition
- Purpose

- Precautions
- Description
- Preparation
- Aftercare
- Risks
- Normal/Abnormal results
- Resources
- Key terms

Special Features

- Over 675 color images including photos, charts, tables, and customized line drawings
- Bold-faced terms function as hyperlinks pointing to related entries
- Cross-references throughout directing readers to related information
- Key terms sections defining unfamiliar concepts
- Contact information for organizations and support groups
- Resources sections with additional sources

Part 3: Medical Content - Sample Entries (C-F Section)

CAFFEINE

Definition: A drug that stimulates the central nervous system

Purpose: Makes people more alert, less drowsy, improves coordination; combined with pain relievers for migraine treatment; used to treat breathing problems in newborns

Sources: Naturally found in coffee, tea, chocolate; colas and soft drinks; available in tablets/capsules (No Doz, Overtime, Pep-Back, Quick-Pep, Caffeine, Vivarin)

Dosage: Adults and children 12+: 100-200 mg every 3-4 hours; timed-release: 200-250 mg once daily

Precautions:

- Cannot replace sleep; can lead to insomnia
- Heavy use builds tolerance and dependence
- Withdrawal symptoms include headaches, fatigue, drowsiness, irritability, vomiting
- Can interfere with sleep if taken near bedtime
- Older people more sensitive to side effects

Special Conditions:

- **Pregnancy:** May cause miscarriage or low birth weight at >300 mg/day
- **Breastfeeding:** Passes into breast milk; babies may become irritable
- **Medical conditions:** Caution with peptic ulcer, heart arrhythmias, high blood pressure, liver disease, anxiety, PMS

Side Effects: Restlessness, nervousness, shakiness, headache, nausea, vomiting; at high doses: anxiety, confusion, heart arrhythmias, convulsions

Interactions: Oral contraceptives, mexiletine, cimetidine, disulfiram affect caffeine breakdown; interferes with iron absorption; dangerous with MAO inhibitors

CALCIUM CHANNEL BLOCKERS

Definition: Medicines that slow calcium movement into heart and blood vessel cells, relaxing blood vessels and reducing heart workload

Purpose: Treat high blood pressure, abnormal heart rhythms, angina pectoris; also used for panic attacks, bipolar disorder, migraine prevention

Common Types: Amlodipine (Norvasc), diltiazem (Cardizem), isradipine (DynaCirc), nifedipine (Adalat, Procardia), nicardipine (Cardene), verapamil (Calan, Isoptin, Verelan)

Precautions:

- Regular physician monitoring required
- Continue taking even when feeling well
- Won't cure high blood pressure but helps control it

- May cause drowsiness; avoid driving until effects known
- May mask chest pain during exercise
- Check pulse regularly as drugs may slow it excessively
- May cause gum swelling and bleeding
- Special Conditions:
- Pregnancy: Large doses caused birth defects in animal studies
- Breastfeeding: Some pass into breast milk
- Medical conditions: May worsen heart/blood vessel disorders; caution with kidney/liver disease, depression, Parkinson's

continue

Part 4: Medical Content - Additional Sample Entries

I'll continue the comprehensive summary of The Gale Encyclopedia of Medicine, Second Edition.

CHRONIC GRANULOMATOUS DISEASE (CGD)

Definition: A genetic disorder affecting the immune system's ability to fight infections

Causes and Symptoms: Genetic defect reduces hydrogen peroxide and superoxide production in white blood cells, preventing them from killing ingested bacteria and fungi; symptoms appear by age two including frequent infections of skin, lungs (pneumonia), mouth (gingivitis), nose, intestines, lymph nodes; liver abscesses and bone infections (osteomyelitis)

Diagnosis: Based on pattern of recurrent infections; blood tests show normal lymphocyte and antibody function but decreased bacterial killing ability

Treatment: Early aggressive antibiotic treatment; immune serum provides antibodies; interferon gamma shows promise as experimental treatment; no cure available

Prognosis: Premature death possible despite treatment, typically from repeated lung infections

Prevention: Cannot be prevented as hereditary; genetic counseling recommended; preventive antibiotics, good hygiene, avoiding crowds can help

CHRONIC KIDNEY FAILURE

Definition: Condition where kidneys gradually lose ability to remove fluids and wastes or maintain proper chemical balance in bloodstream

Description: Affects over 250,000 Americans annually; irreversible and progressive, eventually leading to end-stage renal disease (ESRD); fatal without treatment

Four Most Common Causes:

- Diabetes mellitus: Long-term diabetes damages glomeruli (filtering units)
- Glomerulonephritis: Chronic inflammation of kidney filtering units
- Hypertension: High blood pressure damages kidneys over time; both cause and symptom
- Polycystic kidney disease: Inherited disorder causing cysts on nephrons

Other Causes: Kidney cancer, kidney stones, pyelonephritis, reflux nephropathy, lupus, amyloidosis, sickle cell anemia, Alport syndrome, oxalosis

Symptoms:

- Anemia (kidneys produce erythropoietin for red blood cells)
- Bad breath/taste (urea in saliva)
- Bone/joint problems (vitamin D production affected)
- Edema (swelling around eyes, arms, hands, feet)
- Frequent urination, foamy or bloody urine
- Headaches from hypertension
- Fatigue, itching, lower back pain
- Nausea, vomiting, loss of appetite

Diagnosis: Extensive blood work measuring creatinine, BUN, uric acid, phosphate, sodium, potassium; 24-hour urine collection for protein; imaging (X-ray, MRI, CT, ultrasound) and renal biopsy to determine cause

Treatment:

- Hemodialysis: Most common in US; three times weekly for 3-4 hours; blood circulated through dialyzer (artificial kidney)
- Peritoneal dialysis: Uses peritoneum (abdominal lining) as filter; three types (CAPD, CCPD, IPD)
- Kidney transplantation: From cadaver or living donor; requires immunosuppressive medication
- Dietary management: Low sodium, potassium, phosphorus; protein restrictions; fluid limits
- Medications: EPO for anemia, vitamin supplements, calcium and vitamin D

Prognosis: ESRD lifespan is 18-47% of general population; cadaver transplants have 50% chance of functioning for 9 years; living donor kidneys can function 24+ years

CHRONIC OBSTRUCTIVE LUNG DISEASE (COPD)

Definition: General term for chronic bronchitis and emphysema, characterized by persistent difficulty expelling air from lungs

Description: Nearly 16 million Americans affected (14 million chronic bronchitis, 2 million emphysema); fourth leading cause of death (96,000+ annually); increasing in women due to smoking

Chronic Bronchitis: Inflammation narrows bronchi openings; excessive mucus production; chronic cough with sputum; shortness of breath; damaged cilia increase infection risk

Emphysema: Elastase enzyme overproduction destroys elastin protein in alveoli walls; alveoli rupture and enlarge; lungs become less elastic and overinflated; stale air trapped

Risk Factors:

- Cigarette smoking (80% of cases)
- Cigar and pipe smoking
- Air pollution, industrial dust/fumes
- Age (over 40 for bronchitis, 65+ for emphysema)
- Lower socioeconomic class
- Family clustering/heredity
- Lung infections
- Alpha-1-antitrypsin (AAT) deficiency (early onset emphysema, age 30-40)

Symptoms:

- Chronic bronchitis: Cough, mucus, shortness of breath
- Emphysema: Shortness of breath on exertion, minor cough, little sputum
- Advanced: Respiratory failure, low oxygen, excess carbon dioxide

Diagnosis: Medical history, chest exam; pulmonary function test (spirometry measuring vital capacity and FEV1); chest X-ray; blood gas levels; infection tests; ECG

Treatment:

- Bronchodilators: Three types (beta2 agonists, anticholinergic agents, theophylline)
- Corticosteroids: For chronic bronchitis patients
- Oxygen therapy: Supplemental oxygen from tanks
- Antibiotics: At first sign of respiratory infection
- Vaccines: Annual influenza, one-time pneumococcal
- Expectorants: Loosen mucus
- Diuretics: For heart failure patients
- Augmentation therapy: AAT replacement (Prolastin) for deficiency cases
- Surgery: Rare; lung transplantation or volume reduction

Alternative Treatment: Diet, nutritional supplements, herbal medicines, hydrotherapy, acupressure, aromatherapy, homeopathy, yoga, breathing techniques

Prognosis: Cannot be cured but controlled; median survival ~10 years; related to lung function at diagnosis

Prevention: Quit smoking, avoid irritants/infections, maintain nutrition, proper weight,

exercise, avoid temperature extremes

ELECTROLYTE DISORDERS

Definition: Imbalance of ionized salts (bicarbonate, calcium, chloride, magnesium, phosphate, potassium, sodium) in blood

Description: Electrolytes conduct electricity, balance pH/acid-base levels, facilitate fluid passage (osmosis), regulate neuromuscular/endocrine/excretory systems

Normal Ranges:

- Sodium: 135-145 mEq/L
- Potassium: 3.5-5.5 mEq/L
- Calcium: 8.8-10.4 mg/dL (total); 4.7-5.2 mg/dL (unbound)
- Magnesium: 1.4-2.1 mEq/L
- Chloride: 100-108 mEq/L
- Phosphate: 2.5-4.5 mg/dL
- Bicarbonate: ~27 mM

Causes: Medications, chronic diseases, trauma (burns, fractures), kidney disease, dehydration, excessive intake

Treatment: Address underlying cause; nutritional changes; fluid/electrolyte replacement (IV or oral); hemodialysis for severe cases; medication adjustment

FRACTURES

Definition: Break or disruption in bone tissue or bony cartilage continuity

Types: Simple (closed), compound (open), incomplete (greenstick), complete, comminuted (multiple fragments), compression, impacted, overriding, angulated, displaced, avulsed, segmental

Common Sites: Colles (wrist), Potts (ankle), compound, greenstick (children)

Causes: High activity levels, contact sports, elderly (bone brittleness), diseases (Paget's, rickets, osteogenesis imperfecta, osteoporosis, bone cancer)

Symptoms: Pain increasing with movement, swelling, pale skin, deformity, loss of pulse, numbness, tingling, bleeding/bruising (open fractures)

Diagnosis: Medical history, physical exam, X-ray (multiple views), CT/MRI if needed, tuning fork test for stress fractures, bone scans

Treatment:

- First aid: Proper splinting, blood loss control, vital signs monitoring
- Immobilization: Splints, casts, braces
- Closed reduction: Manual realignment with/without traction
- Open reduction: Surgery with internal fixation (plates, nails, screws, rods)
- Alternative: Electro-stimulation, homeopathy (Arnica, Symphytum), contrast hydrotherapy

Prognosis: Generally good with proper treatment; joint fractures may lead to arthritis; elderly/complicated cases may have non-union

Prevention: Adequate calcium intake, exercise to increase bone density, estrogen replacement for women 50+, safety measures (seat belts, protective gear)

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Part 5: Medical Content - Major Disease Categories

Continuing the comprehensive summary of The Gale Encyclopedia of Medicine, Second Edition.

DIABETES MELLITUS

Definition: Chronic disease where the pancreas no longer produces enough insulin or cells stop responding to insulin, preventing glucose absorption into cells

Statistics: Approximately 14 million Americans (~5% of population); half unaware they have it; chronic disease causing renal failure, heart disease, stroke, blindness

Background:

- Glucose is the body's primary energy source from digested carbohydrates
- Insulin acts like a key opening cell doorways for glucose entry
- Without adequate insulin, glucose remains in blood (hyperglycemia)
- Body attempts to dilute high blood glucose by drawing water from cells, causing excessive thirst and urination
- Cells starve for glucose despite high blood levels

Types:

Type I (Juvenile Diabetes):

- Begins in childhood/adolescence
- Body produces little or no insulin
- Sudden onset, more common in Northern European populations
- Requires daily insulin injections (insulin-dependent)
- ~3 in 1,000 Americans develop Type I
- "Brittle diabetics" have rapid swings between hyperglycemia and hypoglycemia

Type II (Adult-Onset):

- 90%+ of US diabetics
- Occurs in 3-5% under age 50; 10-15% over 50
- Often in overweight, inactive people
- More gradual development than Type I
- May not require insulin initially

Classic Symptoms: Feeling tired/sick, frequent urination, excessive thirst, excessive hunger, weight loss

Complications:

- Ketoacidosis (Type I): Ketones form when body breaks down fats/proteins; symptoms include abdominal pain, vomiting, rapid breathing, lethargy, sweet breath odor; can lead to coma/death
- Heart disease and kidney disease
- Retinopathy (vision problems)
- Neuropathy (nerve damage)
- Poor wound healing
- Birth defects in babies of diabetic mothers

Diagnosis:

- Fasting glucose test: Plasma level ≥ 7 mmol/L indicates diabetes
- Postprandial glucose test: Taken after meal
- Oral glucose tolerance test: Blood samples before/after drinking glucose syrup; ≥ 11.1 mmol/L at 2 hours confirms diabetes
- Home monitoring kits: Finger prick with lancet, analyzed by device

Treatment:

- Diet: 50-60% carbs, 10-20% protein, <30% fat; distributed throughout day; food exchange lists
- Exercise: Weight loss important for Type II
- Oral medications (Type II): Sulfonylureas (tolbutamide, glyburide, glipizide); metformin; acarbose; troglitazone
- Insulin injections (Type I, some Type II): Subcutaneous; types include regular (fast-acting, 15-30 min onset), NPH/Lente (intermediate, 1-3 hr onset), ultra-lente (long-acting, 4-8 hr onset)
- Insulin pumps: Battery-operated, worn externally, programmed for continuous delivery

Prevention: Maintain ideal weight, exercise regularly; avoid triggers like stress, certain medications

DIABETIC KETOACIDOSIS (DKA)

Definition: Life-threatening complication when body breaks down fats instead of glucose for energy, producing acidic ketones

Causes: Most common in Type I diabetes; triggered by interrupted insulin treatment, acute infection, or trauma

Symptoms: High blood sugar (>250 mg/dL, can reach 400-800 mg/dL), frequent urination, thirst, fatigue, nausea, vomiting, abdominal pain, fruity breath odor, rapid deep breathing, muscle stiffness, coma

Diagnosis: Urine/blood strongly positive for glucose and ketones; low serum bicarbonate (<15 mEq/L) indicating acidosis

Treatment (under medical supervision):

- Insulin administration to correct hyperglycemia
- IV fluid replacement
- Electrolyte balancing (prevent hypokalemia)
- Treatment of associated infections

Prognosis: Nearly always successfully treated; mortality rate ~10%; major causes of death are circulatory collapse, hypokalemia, infection, cerebral edema

Prevention: Regular glucose monitoring, insulin administration, lifestyle maintenance; especially important during illness

DIABETIC NEUROPATHY

Definition: Nerve damage throughout body caused by diabetes

Causes: High blood glucose damages nerves and blood vessels supplying nerves; genetic predisposition possible

Types:

- Diffuse peripheral neuropathy: Affects extremities
- Diffuse autonomic neuropathy: Affects internal organs
- Focal neuropathy: Affects specific nerves

Symptoms by Type:

Peripheral: Numbness, tingling, burning, insensitivity to pain, needle-like jabs, extreme touch sensitivity, loss of balance/coordination

Autonomic: Impaired urination/sexual function, bladder infections, stomach disorders (gastric stasis), nausea, vomiting, bloating, dizziness, fainting, appetite loss

Focal: Thigh pain, severe lower back pain, chest/stomach pain, eye pain, double vision, facial paralysis

Severe Cases: Loss of sensation leads to unnoticed injuries, progressing to infections, ulceration, amputation

Diagnosis:

- Physical exam and pain assessment
- Screening test for lost sensation
- Nerve conduction studies
- Electromyography (EMG)
- Ultrasound
- Sometimes nerve biopsy

Specialists: Neurologists, urologists, gastroenterologists, podiatrists

Treatment:

- Improve blood sugar control (may heal early stages)
- Pain management: Capsaicin cream, tricyclic antidepressants (imipramine, nortriptyline), narcotic analgesics (morphine), anticonvulsants (carbamazepine, gabapentin), antiarrhythmics

Prognosis: Early stage can be reversed with good glucose control; established nerve damage cannot be reversed; recovery is slow if it occurs

Prevention: Tight glucose control, avoid alcohol and cigarettes

DIABETIC FOOT INFECTIONS

Definition: Infections in skin, muscles, or bones of foot due to nerve damage and poor circulation from diabetes

Risk Factors:

- Age >60
- Poorly controlled diabetes

- Foot ulcers, laser treatment for retina changes
- Kidney/vascular disease
- Loss of sensation (neuropathy)

Causes: Bacteria enter through small cracks/fissures, scar tissue, foot ulcers from pressure/nerve damage/poor circulation, injuries from poorly fitting shoes

Common Sites: Ball of foot, bottom of big toe, sides of foot

Diagnosis: X-ray to check bone infection; wound culture to identify organism; antibiotic selection based on culture results

Treatment:

 - Appropriate antibiotic based on culture
 - Surgical removal of dead/infected tissue
 - Cast/special shoes for protection
 - Stay off feet
 - Good diabetes control
 - Surgery to increase blood flow if ulcer doesn't heal

Alternative Treatment: Acupuncture, vitamin C (boost infection-fighting), other vitamins/herbs for general health

Prognosis: Without proper treatment, can lead to serious illness, gangrene, amputation, death; with proper treatment and good foot care, excellent outcomes

Prevention: Regular foot inspection, proper footwear, good diabetes control

DIABETES INSIPIDUS (DI)

Definition: Disorder involving inappropriate antidiuretic hormone (ADH) function; unrelated to diabetes mellitus

Types:

- Central DI: Insufficient ADH production by pituitary
 - Nephrogenic DI: Kidneys can't respond to ADH properly
- Mechanism: ADH normally regulates fluid retention by kidneys based on plasma osmolality (solute-to-solvent concentration)

Statistics: Average onset age 24; more common in males

Causes - Central DI:

- May run in families
- Head injury
- Brain surgery
- Cancers spreading to pituitary (especially breast cancer)
- Sarcoidosis
- Decreased oxygen delivery to brain
- Certain medications (phenytoin)
- Excessive alcohol use
- Pregnancy/postpartum
- AIDS with brain infections

Causes - Nephrogenic DI: Various kidney disorders

Symptoms: Extraordinarily large urine output (10+ liters/day), excessive thirst, highly concentrated blood with low fluid volume

Diagnosis: Based on pattern of symptoms; distinguishing between central DI, nephrogenic DI, partial DI, or psychogenic polydipsia

Treatment:

- Medications: Vasopressin (Pitressin) injected, desmopressin acetate (DDAVP) inhaled, antidiuretic drugs (chlorpropamide, clofibrate, carbamazepine)
- Nephrogenic DI: Special diets restricting solute intake, thiazide diuretics

Prognosis: Controllable with adequate water intake; most patients lead normal lives

CRYPTOCOCCOSIS

Definition: Fungal infection caused by inhaling *Cryptococcus neoformans*; most common fungal infection in AIDS patients

Description: Fungus found worldwide in soil contaminated with bird droppings, especially pigeon; rare in healthy individuals; also called cryptococcal meningitis, Busse-Buschke disease, European blastomycosis, torular meningitis, torulosis
Infection Progression:

1. Immune system heals without intervention (asymptomatic)
2. Disease stays localized in lungs
3. Spreads throughout body (opportunistic infection)

Statistics: 6-10% of US AIDS patients develop cryptococcosis

Symptoms - Early/Lung: Dry cough, respiratory infection symptoms, nodules visible on chest X-ray

Symptoms - Meningitis (most common presentation):

- Fever and headache (85% of patients)
- Nausea, vomiting, weight loss, fatigue
- Blurred vision, stiff neck, light aversion, seizures (20-30%)
- Brain inflammation (encephalitis)
- Brain lesions (cryptococcomas/tortulomas)

Other Sites: Kidneys, bone marrow, heart, adrenal glands, lymph nodes, urinary tract, blood, skin (painless rashes mimicking molluscum contagiosum)

Diagnosis:

- Blood and CSF antigen tests (99% accurate for blood test)
- India ink test on CSF (75-85% accurate)
- Culture of sputum, tissue, or CSF
- Lumbar puncture for CSF sample
- Chest X-rays

Treatment:

- Initial: Amphotericin B (Fungizone) ± 5-flucytosine (Ancobon) IV for 2-3 weeks
- Maintenance: Oral fluconazole (Diflucan) for life in AIDS patients
- Preventative: Fluconazole for advanced AIDS patients
- Pfizer financial assistance program for fluconazole

Prognosis: Untreated is fatal; acute mortality rate 10-25% in AIDS patients; most deaths within 2 weeks of diagnosis; 50-60% relapse within 6 months without suppressive therapy; neurologic symptoms may remain (weakness, headache, hearing/visual loss, hydrocephalus)

Prevention: Avoid HIV infection; stay away from bird droppings in contaminated areas

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